

Quiz 3

Name: _____

1. Reflections. Let $f(x) = \sqrt{x}$.

(a) Reflect the graph of $y = f(x)$ across the y -axis. Write the new function.

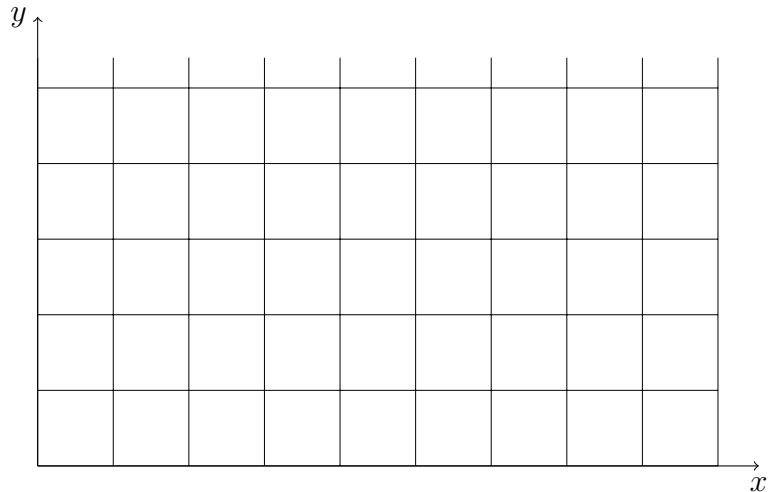
(b) Then reflect that new graph across the x -axis. Write the final function.

2. Average Rate of Change / Linearity.

The table shows time x (hours) and population y (grams).

(a) Plot the ordered pairs (x, y) on the coordinate grid below.

Time x	Population y
0	0.09
1	0.12
2	0.16
3	0.22
4	0.29
5	0.39



- (b) Compute the average rate of change $m = \Delta y / \Delta x$ between each pair of consecutive points (from $x = 0 \rightarrow 1$, $1 \rightarrow 2$, $\dots$, $4 \rightarrow 5$). Is the function linear? Explain briefly.

3. Intercepts and Zero. Let $f(x) = -\frac{1}{2}x + 2$.

- (a) Find the y -intercept.
- (b) Find the zero of f (equivalently, the x -intercept).

End of quiz.